The Los Alamos HIV Molecular Immunology Database

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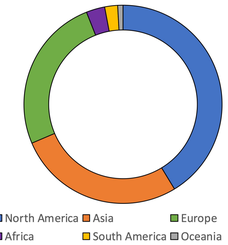
HISTORY OF DATABASE:

- LANL's HIV Database was the first host-pathogen-specific database
- The HIV (pathogen) Sequence Database was founded in 1986
- The HIV Molecular Immunology Database (host response) was founded in 1994 by Bette Korber

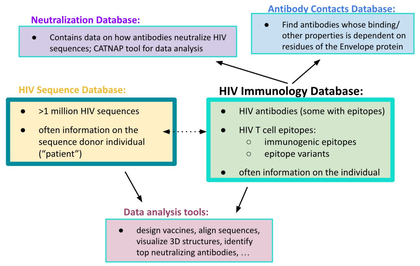
DATABASE USAGE:

- LANL's HIV Databases are used by researchers across the world:

Database visits by continent:

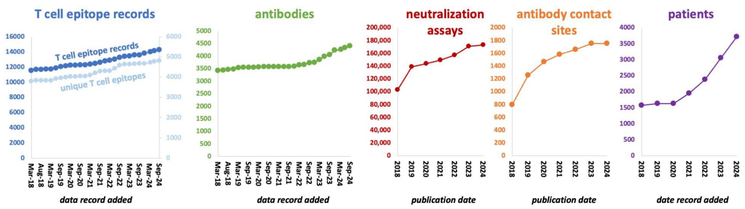


- Organization of LANL's HIV Databases:



- Entries are continuously being added to our database:

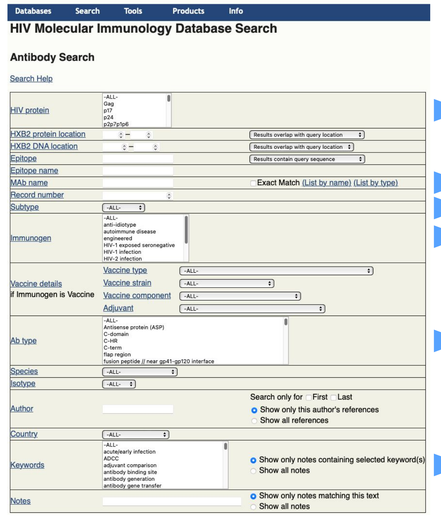
Number of entries in the HIV Molecular Immunology Database



RECENT UPDATES:

- Outreach Activities: we will host a HIV Databases User Workshop at Keystone 2025's "HIV Cure" conference
- Our HIV Genome Browser Tool has been updated:
  - Tracks can be filtered (e.g. by antibody name or HLA allele)
  - A track exists for antibody contact sites on Env
  - Tracks display immunogenicity of T cell epitopes
- Our Env Immunogen Browser is now live
- Our CAByN (Choosing Antibodies by Neutralization) tool is now live
- Our HIV 3D Viewer tool has been updated

ANTIBODY SEARCHES:

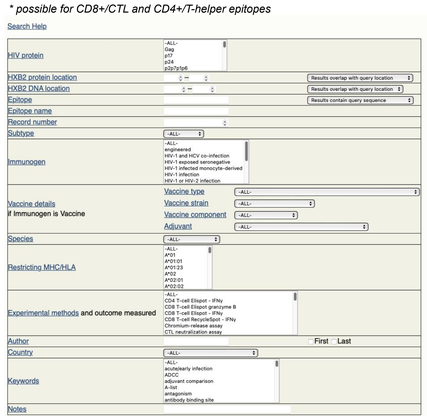


examples of search options:

- HIV protein *(e.g. Env)*
- antibody name
- HIV viral subtype
- type of immunogen *(e.g. vaccine)*
- type of antibody *(e.g. gp41 MPER)*
- keywords *(e.g. effector function, escape)*

T CELL EPITOPE SEARCHES:

** possible for CD8+/CTL and CD4+/T-helper epitopes*

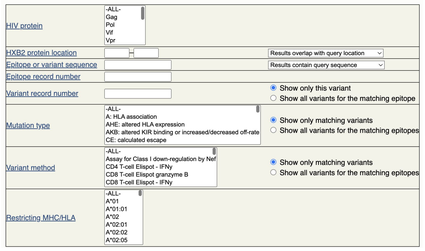


examples of search options:

- epitope *(e.g. GELDRWEKI)* *"contain", "exact", or "fuzzy" match options*
- immunogen *(e.g. HIV-1 infection)*
- restricting HLA *(e.g. HLA-A*02:01)*
- experimental methods & outcome *(e.g. tetramer binding)*
- keywords *(e.g. immunodominance)*

T CELL VARIANT EPITOPE SEARCHES:

** possible for CD8+/CTL and CD4+/T-helper epitope variants*

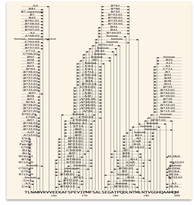


examples of search options:

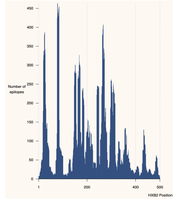
- epitope or variant sequence
- mutation type *(e.g. drug induced)*
- variant method *(e.g. cytokine production)*

MAPS, PLOTS, & TABLES::

- Our database has interactive maps of each HIV protein with immune epitopes overlaid
- This information is also available as epitope density plots for each HIV protein



Gag Interactive map with clickable CTL/CD8+ HIV Epitope links



Gag CTL/CD8+ HIV Epitope Density Plot

- We provide a table of best-defined CD8+/**CTL epitopes** ("A-list"), which is maintained by Christian Brander
- We also provide a table of **ADCC** (antibody-dependent cellular cytotoxic) **antibodies**

SOME OF OUR TOOLS:

HIV; custom input
HIV-specific

- **CATNAP** (Compile, Analyze and Tally Neutralizing Antibody Panels):
  - Displays antibody neutralization data for selected viruses
  - Calculates viral features associated with neutralization activity
  - Data is downloadable
- **CombiNAber:**
  - Predicts optimal neutralizing combinations of up to 4 antibodies
- **Epigraph Vaccine Suite:**
  - Designs T-cell epitope vaccines with optimal population coverage
- **HIV Genome Browser:**
  - Interactive; displays location of epitopes and antibody contacts
- **CAByN** (Choose antibodies by neutralization):
  - Generates a list of neutralizing antibodies based on CATNAP neutralization data, from user-input select criteria
- **HIV 3D Viewer:**
  - Displays location of selected residues in a 3D structure view of an HIV protein

UPCOMING IMPROVEMENTS:

- We will soon allow users to search for antibody effector functions
  - ADCC (antibody-dependent cell-mediated cytotoxicity) results will be categorized by type of assay

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